

LUMBERTON, NC, USA

WMO: 723108

Lat: 34.608N

Lon: 79.059W

Elev: 37

StdP: 100.88

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13776

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.3	-14.5	1.1	-1.4	-12.1	1.3	1.1	10.6	9.2	9.2	8.7	1.9	310	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.2	24.7	33.9	24.4	32.7	24.0	26.4	32.2	25.8	31.2	25.3	30.5	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.0	28.6	24.2	19.2	27.8	23.9	18.8	27.5	82.0	32.2	79.6	31.5	77.3	30.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.7	-9.1	37.4	2.0	1.6	-10.5	38.5	-11.7	39.4	-12.8	40.3	-14.2	41.4		
(5)			-10.0	27.4	1.8	0.7	-11.4	27.9	-12.4	28.3	-13.5	28.8	-14.8	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	6.8	8.5	12.3	16.9	21.6	25.9	27.1	26.6	23.4	17.6	12.2	8.6		
	DBStd	8.36	5.82	5.07	5.14	4.32	3.56	2.54	1.95	2.15	2.87	4.45	4.78	5.34		
	HDD10.0	383	134	83	37	3	0	0	0	0	0	2	30	93		
	HDD18.3	1488	358	278	196	77	13	0	0	0	3	67	191	305		
	CDD10.0	3061	36	41	108	209	359	478	530	513	403	239	96	49		
	CDD18.3	1123	1	2	9	33	113	228	272	255	155	46	7	3		
	CDH23.3	10322	2	16	77	321	1054	2202	2691	2351	1191	363	44	10		
(13)	CDH26.7	4243	0	1	7	66	383	1006	1232	1037	427	81	3	1		
(14)	Wind	WSAvg	2.7	2.9	3.1	3.3	3.2	2.8	2.5	2.4	2.1	2.5	2.4	2.6		
(15) Precipitation	PrecAvg	1371	110	99	115	82	110	141	166	152	126	89	82	86		
	PrecMax	1752	218	204	284	204	258	371	334	317	430	270	192	207		
	PrecMin	1032	7	22	12	3	26	54	38	27	8	0	20	4		
	PrecStd	189	52	49	67	44	58	64	65	69	86	61	45	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	25.9	27.7	30.9	33.9	37.1	36.6	37.1	34.0	31.0	27.1	24.6		
		MCWB	18.0	18.6	18.2	20.9	22.9	24.4	25.2	24.6	22.9	22.7	20.1	20.0		
	2%	DB	21.1	22.6	25.7	28.6	32.2	34.9	34.9	34.5	32.5	28.8	24.7	22.4		
		MCWB	16.5	16.6	17.7	19.9	22.0	24.2	25.1	25.0	23.1	21.0	18.4	18.5		
	5%	DB	18.8	20.0	23.3	27.2	30.8	33.3	33.7	33.0	31.0	27.3	22.7	20.2		
		MCWB	15.2	15.1	16.0	19.0	21.5	23.6	24.9	24.8	22.8	20.1	17.1	17.4		
	10%	DB	16.7	17.6	21.3	25.2	28.8	32.1	32.4	32.0	29.1	25.3	21.0	17.8		
		MCWB	13.4	13.4	15.3	18.0	20.8	23.3	24.6	24.6	22.2	19.7	16.8	15.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.7	20.1	23.0	24.4	26.2	27.2	27.2	25.7	24.6	21.8	21.3		
		MCDB	21.6	23.4	24.6	27.7	30.7	32.9	33.4	32.2	30.3	27.1	24.7	23.4		
	2%	WB	17.7	18.0	19.0	21.6	23.5	25.4	26.5	26.4	24.9	23.5	20.2	19.3		
		MCDB	20.1	21.3	23.2	26.1	29.3	31.8	32.5	31.9	29.2	26.3	22.7	21.1		
	5%	WB	16.0	16.4	17.9	20.4	22.7	24.9	25.8	25.8	24.2	22.4	19.0	17.7		
		MCDB	18.2	19.1	21.5	25.0	27.9	30.7	31.4	30.9	28.2	25.3	21.5	19.7		
	10%	WB	13.9	14.1	16.8	19.3	21.9	24.3	25.4	25.3	23.6	21.1	17.5	15.8		
		MCDB	16.2	17.2	20.2	23.6	26.5	29.5	30.6	30.2	27.0	23.9	19.9	18.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.8	12.5	13.0	12.5	11.0	10.2	10.0	10.4	12.5	13.2	11.2		
		MCDBR	13.2	14.3	14.3	14.3	14.1	12.7	11.6	11.7	12.4	13.7	14.5	12.6		
		MCWBR	9.2	9.0	7.6	6.8	5.4	4.0	3.8	4.0	4.7	6.5	8.5	9.1		
	5% WB	MCDBR	11.5	13.2	11.8	11.7	11.3	10.7	10.4	10.1	9.3	9.4	10.9	11.3		
		MCWBR	9.0	9.7	7.6	6.4	4.8	3.8	3.9	3.9	4.1	5.2	7.7	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.330	0.368	0.404	0.466	0.495	0.541	0.515	0.431	0.395	0.344	0.319		
	taud		2.490	2.459	2.381	2.276	2.153	2.113	2.004	2.067	2.299	2.364	2.466	2.531		
	Ebn at Noon		886	916	904	883	829	801	761	774	829	831	850	868		
	Edh at Noon		85	97	114	133	152	158	176	162	123	106	86	77		
(44) All-Sky Solar Radiation	RadAvg		2.71	3.36	4.45	5.57	6.08	6.36	6.07	5.46	4.66	3.86	2.97	2.37		
	RadStd		0.16	0.35	0.34	0.37	0.48	0.41	0.38	0.27	0.38	0.46	0.24	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (64 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+120	

Nomenclature: See separate page